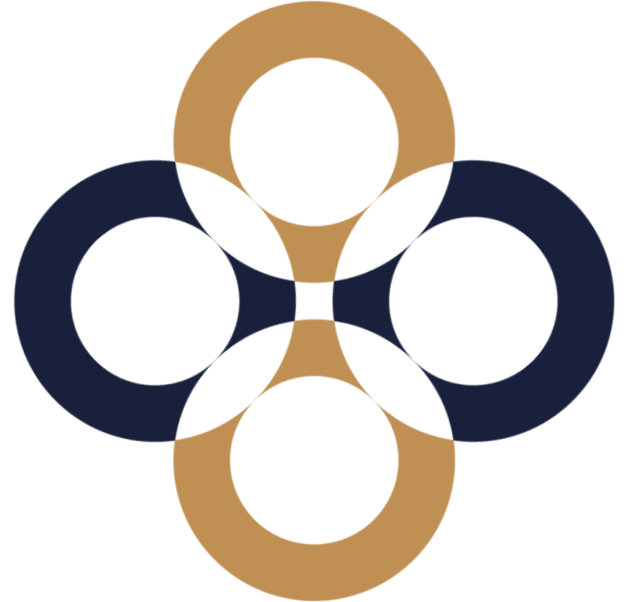
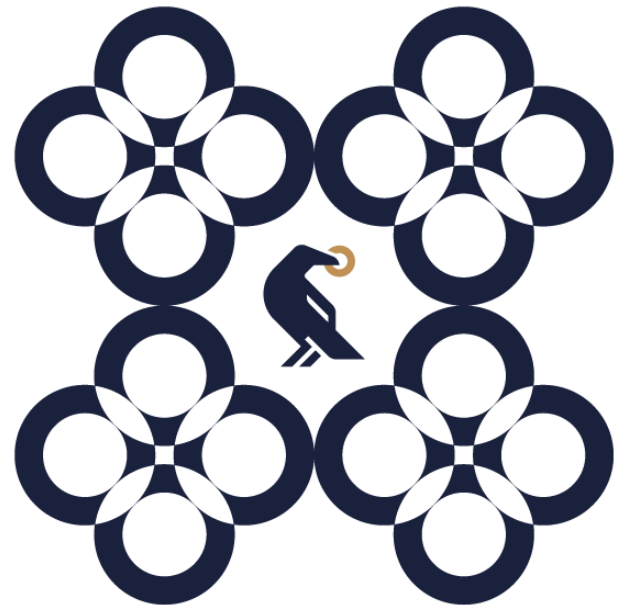


Special challenges of DCF-based valuation

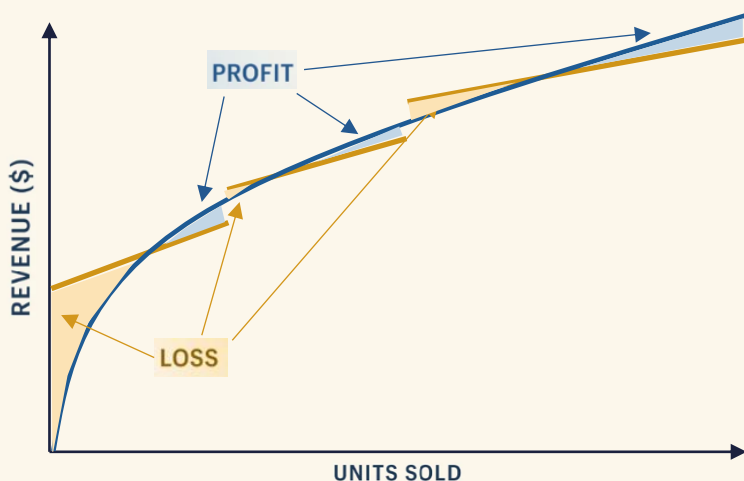
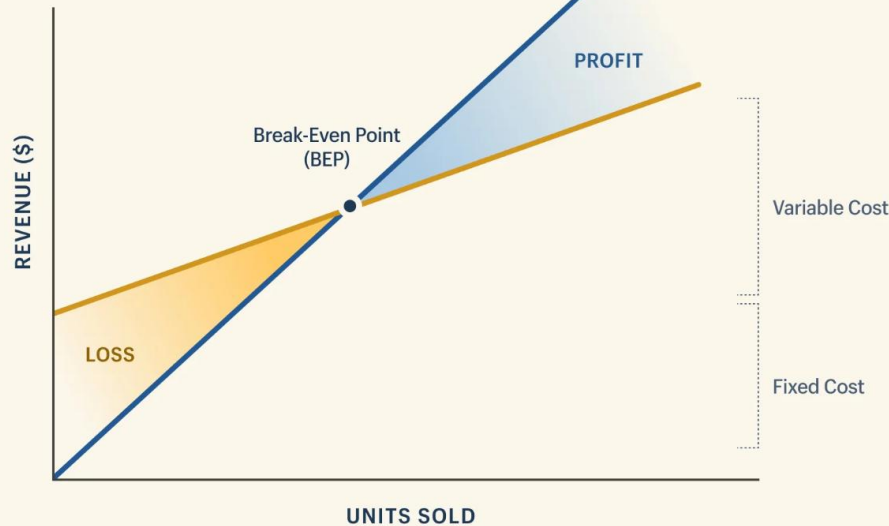
Péter Juhász, PhD, CFA



Modelling & Simulation

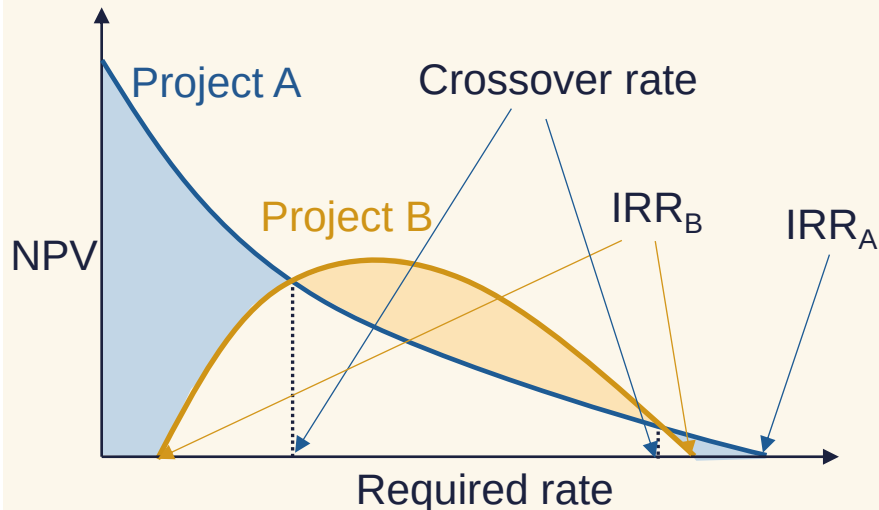
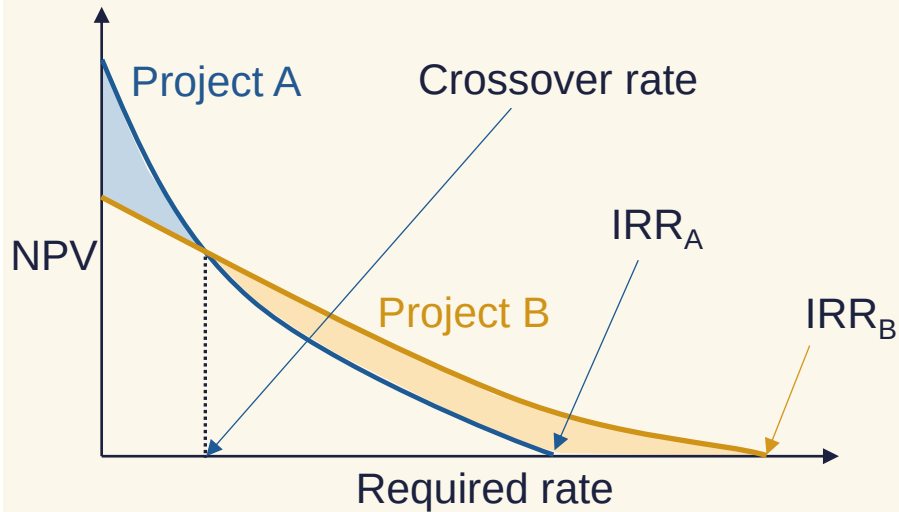


Break-even analysis



1. For what volumes would the operation be profitable?
2. Contrasts additional revenues and expenses
3. Focuses on a given period only
4. In the long run all costs are variable
5. At what price would we take the order?

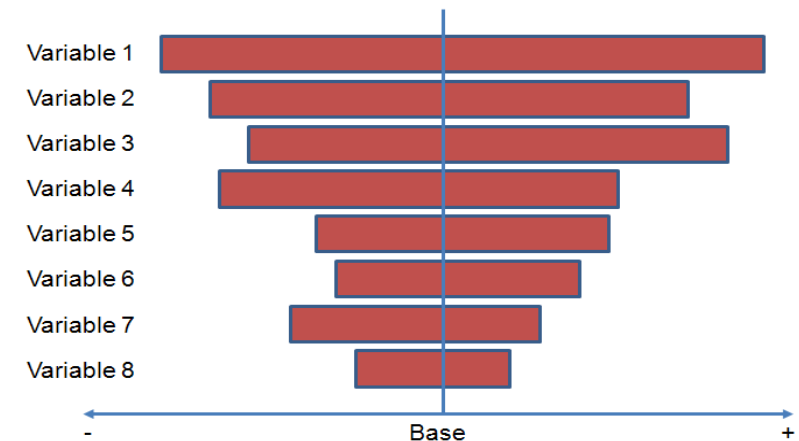
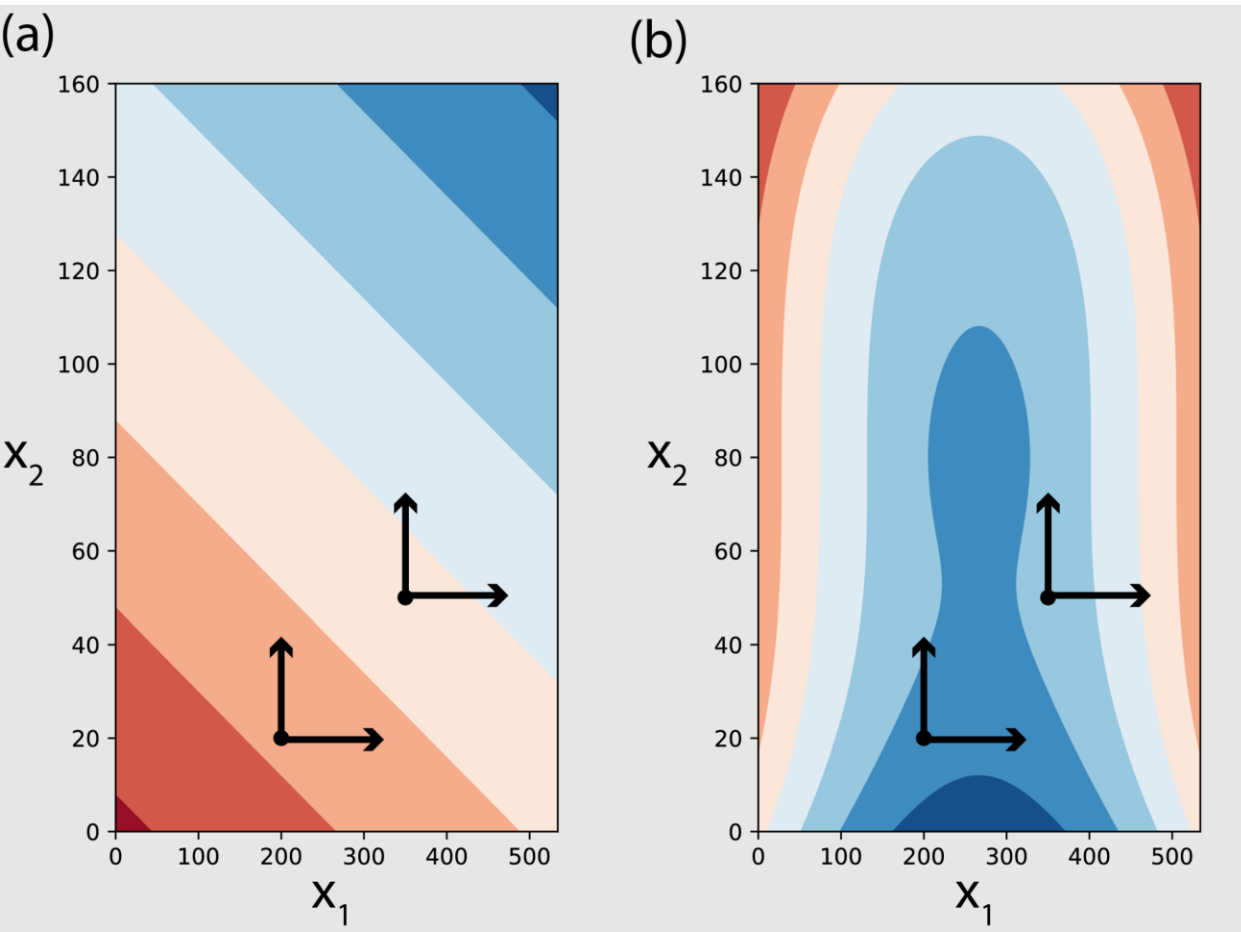
Crossover rate analysis



1. Contrasting project NPVs at multiple required rates
2. If the CF distribution across time differs the reaction of NPVs differ
3. Visualises the risk in cost of capital estimation
4. Does not consider project length

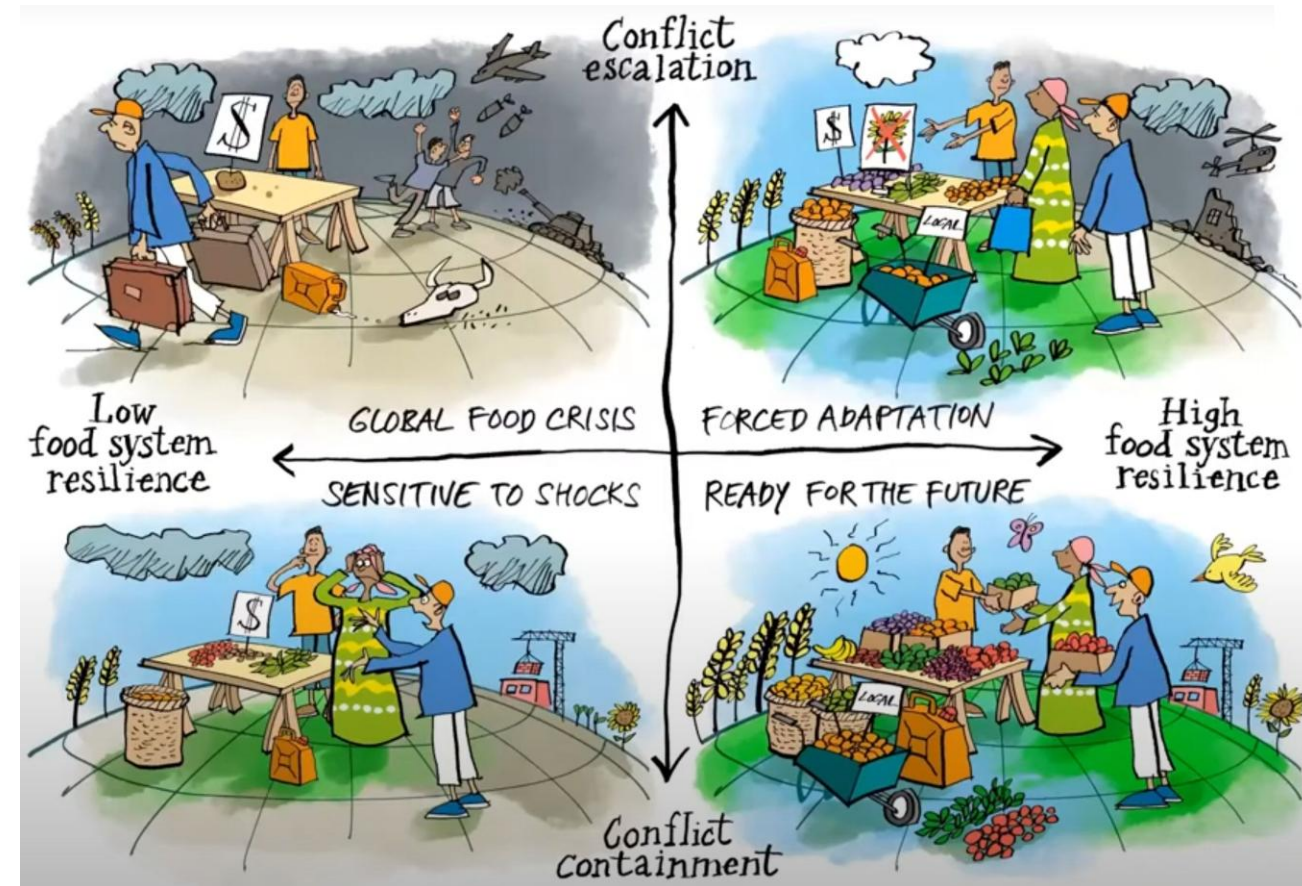
Sensitivity analysis

1. Substitutes expected values by the extremes of the likely value range
2. Identifies key variables to be estimated at highest precision



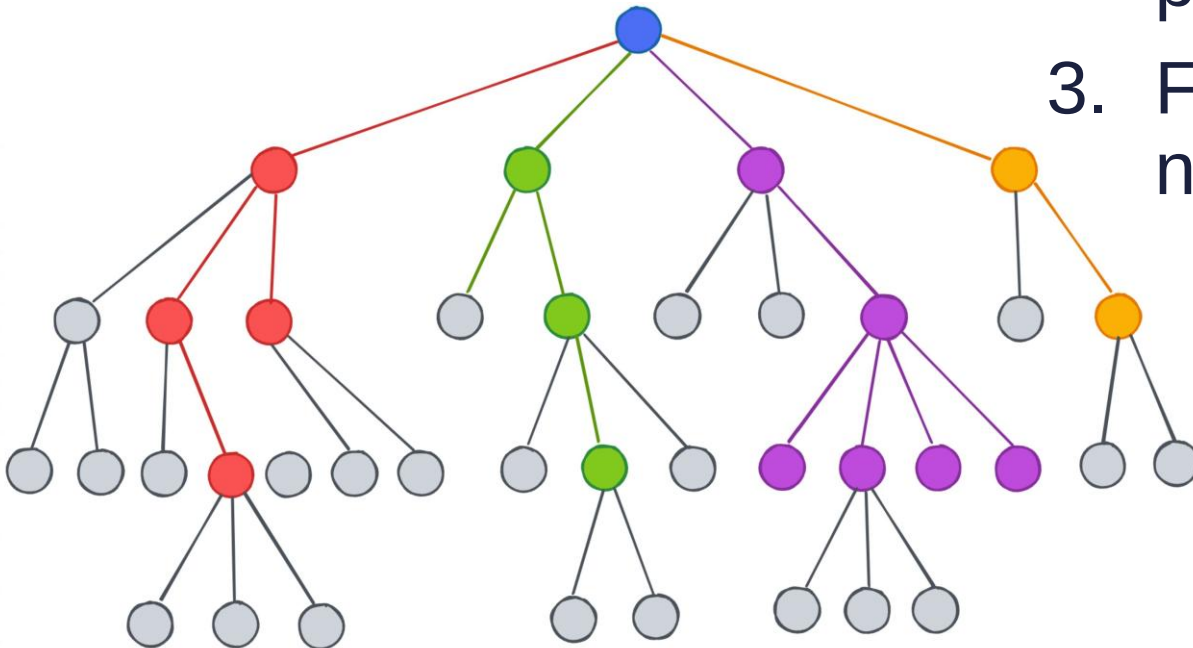
Scenario analysis

1. List possible believable future states of world.
2. Pick the 3-5 most likely of those and describe them in detail.
3. Choose the best strategy for each of the scenarios.
4. Weight the outcomes for each scenario by the probability of the scenario to happen.



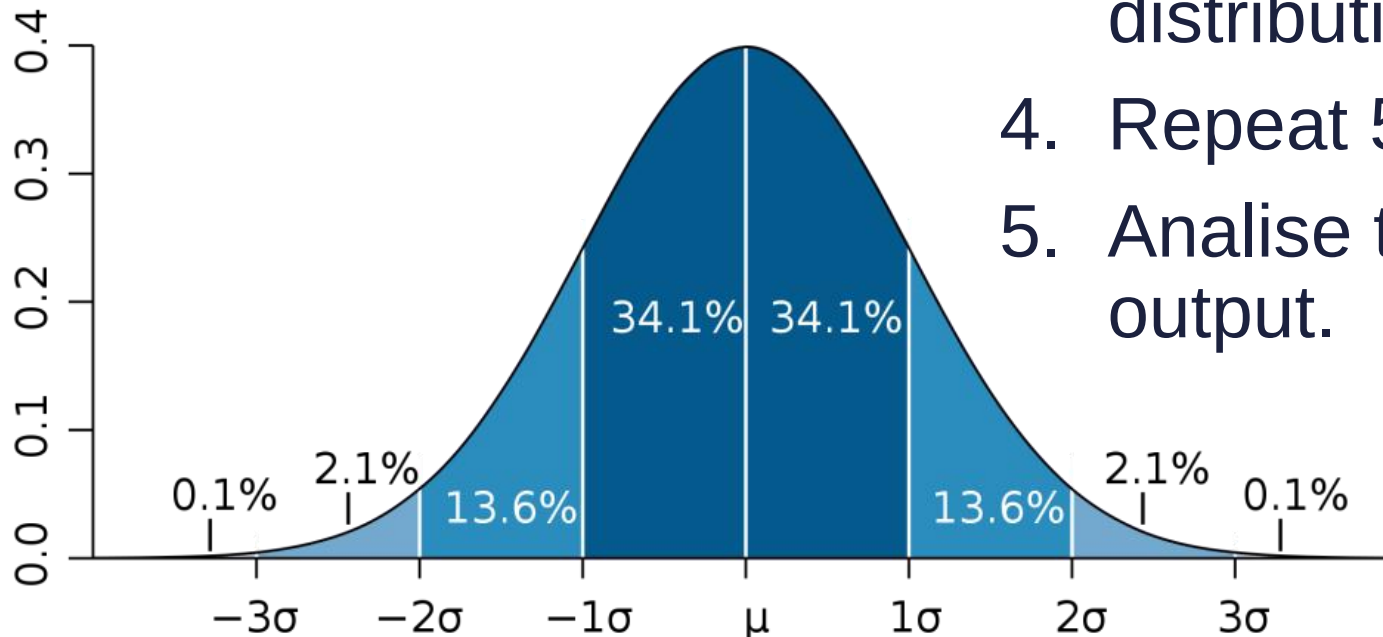
Decision trees

1. Find the decision points for the given scenario.
2. Describe choices of the decisionmaker (could be the nature!). List outcomes, probabilities, consequences.
3. Find the best choice for each of the nodes backwards.



Monte Carlo simulation

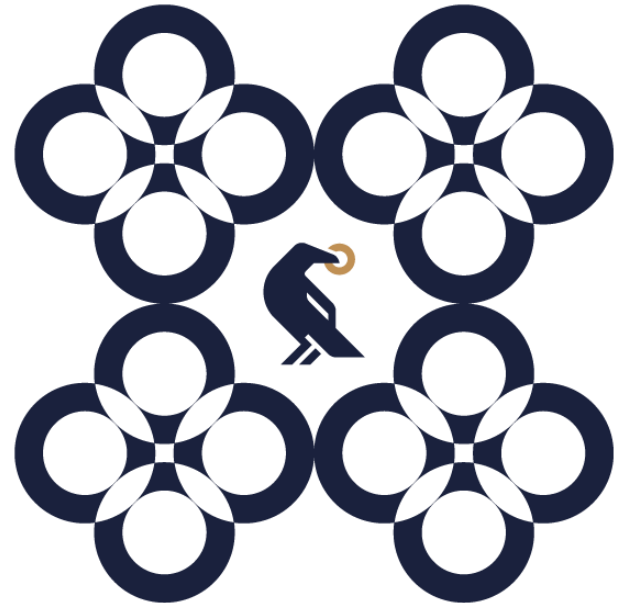
1. Find the key input parameters.
2. Describe the distribution of those inputs.
3. For each of the variable pick a random realisation from the distribution.
4. Repeat 5,000 to 100,000 times.
5. Analyse the distribution of the model output.



Opportunities that gain value for an increase in risk

Poor projects are often offered for sale as real options

Multi-business firms



Multi-business companies

Business units have different risks so CFs can not be merged

Invested capital and required rate for Bus can often be estimated only.

We must address the transfer prices

What would be the market price?

HQ may have a negative value

Is there any synergy? Where does it come from?

Valuing a multi-BU firm 1

Perpetuity-like FCFF=100, $g=5\%$, $D/V=50\%$, $r_D=5\%$ $t=20\%$,
 $r_A=10,5\%$, thus: WACC=10%.

Business value:

$$V = \frac{FCFF}{(WACC - g)} = \frac{100}{(10\% - 5\%)} = 2000$$

Valuing a multi-BU firm 2

BU A FCFF=30, BU B FCFF=90, HQ FCFF=-20.

A superficial estimation:

$$V_A = \frac{30}{(10\% - 5\%)} = 600$$

$$V_B = \frac{90}{(10\% - 5\%)} = 1800$$

$$V_K = \frac{-20}{(10\% - 5\%)} = -400$$



Valuing a multi-BU firm 3

Detailed analysis:

	A	B	HQ	Average*
FCFF	30.00	90.00	-20.00	100.00
g	7.00%	4.33%	5.00%	5.00%
r_A	10.00%	10.66%	10.50%	10.50%
WACC	9.50%	10.16%	10.00%	10.00%

*FCFFs used as weights.

Valuing a multi-BU firm 4

Enhanced valuation:

$$V_A = \frac{30}{(9,5\% - 7\%)} = 1200,00$$

$$V_B = \frac{90}{(10,16\% - 4,33\%)} = 1543,74$$

$$V_K = \frac{-20}{(10\% - 5\%)} = -400,00$$

$$\mathbf{V=2343.74}$$

Valuing a multi-BU firm 5

If BU A would be a stand-alone firm, its FCFF were 20, while that of the HWQ -15 (synergy lost):

$$V_A = \frac{20}{(9,5\% - 7\%)} = 800,00$$

$$V_B = \frac{90}{(10,16\% - 4,33\%)} = 1543,74$$

$$V_K = \frac{-15}{(10\% - 5\%)} = -300,00$$

Valuing a multi-BU firm 6

At what price would we sell BU A?

$$V_A = V_{A+B+K} - V_{B+K}$$

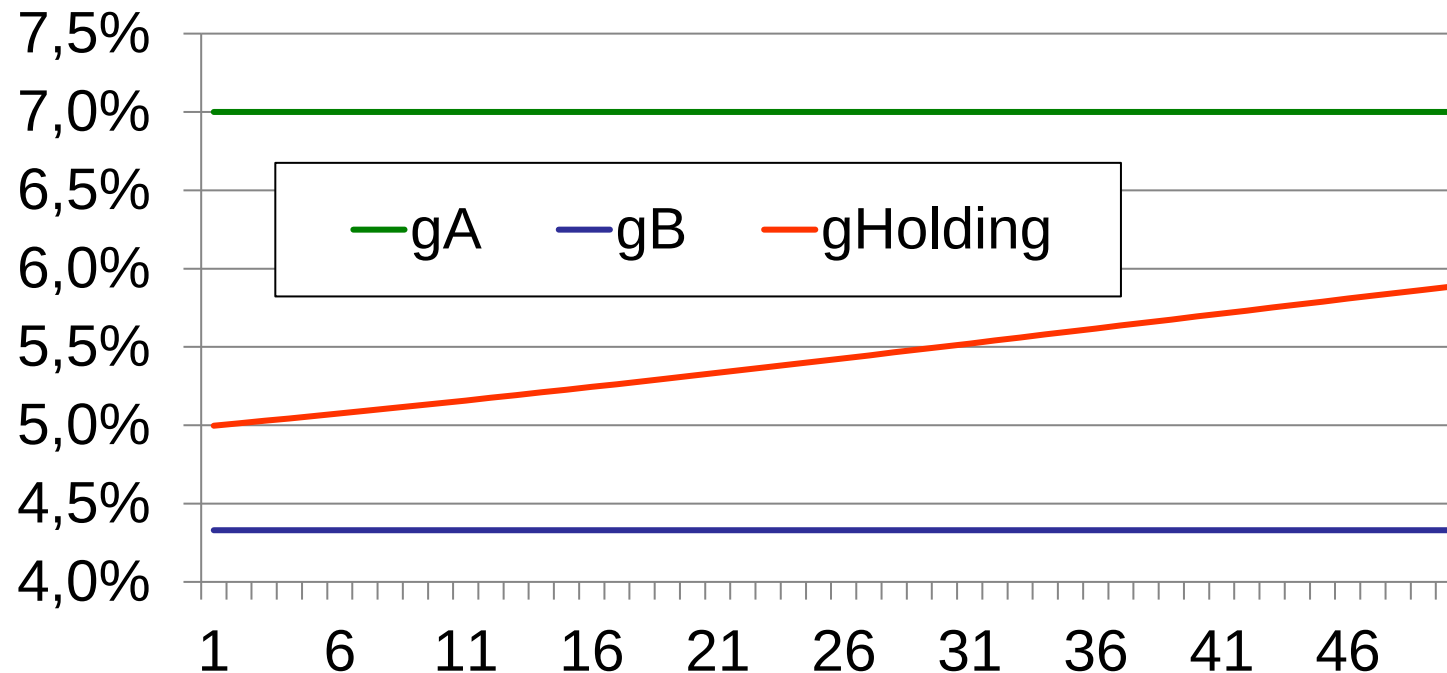
We do not need the value of BU A.

$$V_A = 2343,74 - (1543,74 - 300) = 1100$$

This is the value of the BU decreased by the costs emerging at the HQ.

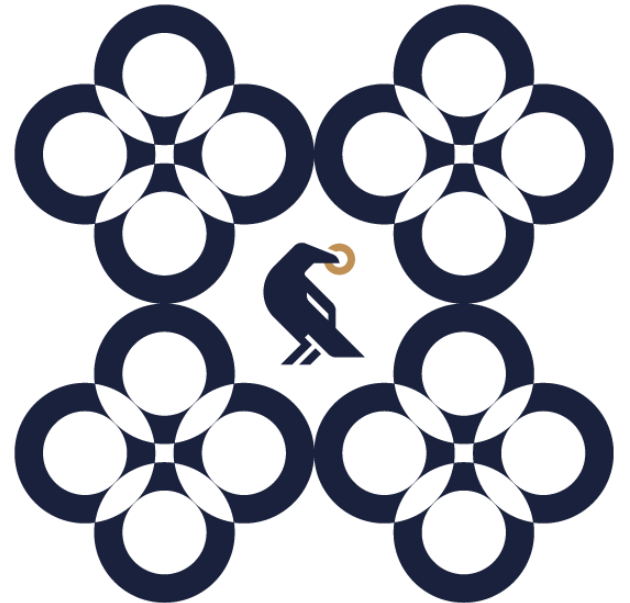
Valuing a multi-BU firm 7

What is the actual growth rate of the firm if there would be no change in the operation?



Perpetuity-like BUs do not necessarily add up to a perpetuity.

Inflation & FX rates



Effects of the inflation

Inflation is not value neutral.

1. Not all items increase with inflation: D&A
2. Not all items are driven by the same inflation: selling price, procurement price, energy, wages, import, export
3. Effect of inflation may appear delayed: inventory



**Valuation may only be done
based on nominal plans.**

Exchange rate risk

FX rate changes due to inflation rate differences are not to be managed; only purchase parity deviations should be addressed

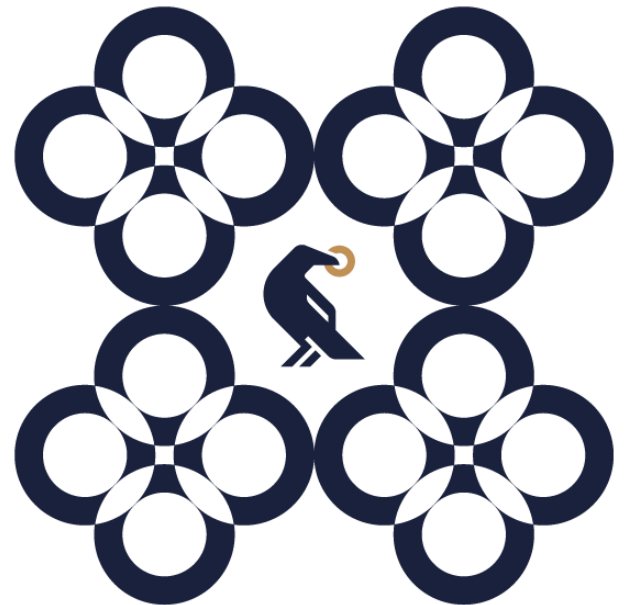
Market hedging
(forward, swap,
option)

A collection of wooden currency symbols including the Euro (€), Dollar (\$), Pound (£), Yen (¥), and others, scattered on a brown surface.

Corporate
diversification
across markets

Investor
diversification

Seasonality

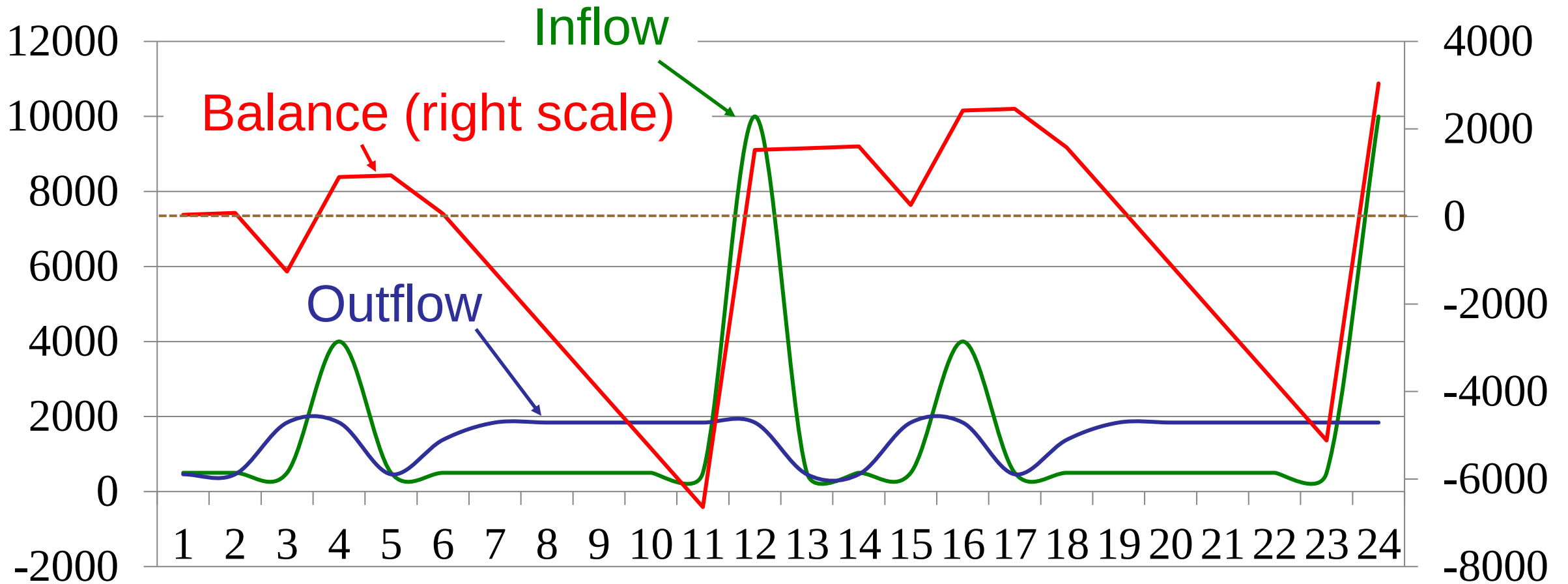


Are all CFs due
at the same
point in time?

Is that point the
end of the
year?

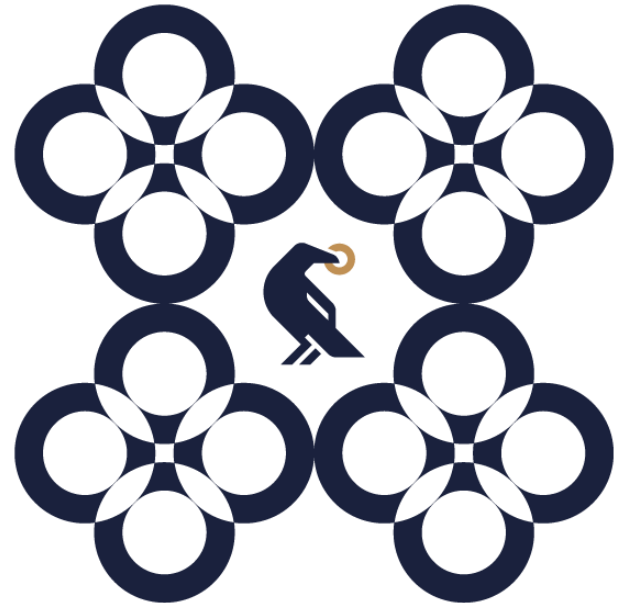


Seasonality effect on cash flows



At a rare of 1% the multiplier is 87.32%.

Valuing liabilities



Valuing liabilities

Bank loans

Bonds

Rental
agreements

Lease

Convertible bonds

Management
options

Minority interests

Preferred equity

Pension funds

Legal processes

Environmental
obligations

Generally, at book value (if a variable rate debt)

Fixed rate debt and subsidised debt calls for a separate valuation (May influence WACC calculation!)

In case of financial difficulties, the value of debt is directly linked to the value of the firm

Non-operational debt should not be considered in WACC

Off-balance sheet items

E.g. legal processes: extent, probability, date

Dividend preferred shares

Like uncovered debt

Options

Management options, warrants, convertible debt

Minority interests

Needs valuation of affiliates

Limits of applying the DCF approach 1

Loss-making firms

The available information is not relevant

- Negative CF
- Growth rate is hard to estimate
- Tax calculation is challenging
- Continuous operations is not realistic (bankruptcy)



Limits of applying the DCF approach 2

Young or new companies

the necessary information is not existing

Probabilities only $\Rightarrow$ Venture capital

Differences: frequency of information service,
required rate depends on the VC firm

Liquidity discount, extra risk due to fast growth



Limits of applying the DCF approach 3

Private firms

the necessary information is not available

More lax reporting rules than elsewhere
(comparability?)

Less information shared
(Auditor? Analysts?)

Exit is harder

No market data (competitors data, betas)

Management and ownership overlapping
(salary vs dividend, non-operational assets)



A DCF „alternatives”

1

Use non-CF data
(Dividend, net profit)

2

Use data from other firms
(Multiple-based methods)

3

Use alternative approaches
(going concern vs recreation or liquidation)

**Thank you
for your attention.
Any questions?**

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